

TF-IDF Features

Step-by-Step Manual Calculations from Scikit-learn

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Example Dataset

We use four short documents for binary spam classification:

Document	Text	Label
D_1	free money click now	1
D_2	meeting tomorrow at university	0
D_3	claim your free prize	1
D_4	please submit the assignment	0

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(texts)
model = LogisticRegression()
model.fit(X, labels)
```

Step 1: Vocabulary Construction

`TfidfVectorizer` tokenizes the documents and sorts the vocabulary alphabetically.

Index	Feature	Index	Feature
0	assignment	8	please
1	at	9	prize
2	claim	10	submit
3	click	11	the
4	free	12	tomorrow
5	meeting	13	university
6	money	14	your
7	now		

There are 15 unique terms, so each document becomes a vector in $\mathbb{R}^{15}$.

Step 2: Document Frequency

The document frequency $df(t)$ counts how many documents contain term t .

Term	Documents	df	Term	Documents	df
assignment	D_4	1	please	D_4	1
at	D_2	1	prize	D_3	1
claim	D_3	1	submit	D_4	1
click	D_1	1	the	D_4	1
free	D_1, D_3	2	tomorrow	D_2	1
meeting	D_2	1	university	D_2	1
money	D_1	1	your	D_3	1
now	D_1	1			

Only free appears in more than one document.

Step 3: Scikit-learn IDF Formula

With default `smooth_idf=True`, scikit-learn uses

$$\text{IDF}(t) = \log \left(\frac{1 + N}{1 + df(t)} \right) + 1,$$

where $N = 4$ documents.

For a term appearing once:

$$\text{IDF} = \log \left(\frac{5}{2} \right) + 1 = 0.9162907 + 1 = \boxed{1.9162907}.$$

For `free`, which appears twice:

$$\text{IDF}(\text{free}) = \log \left(\frac{5}{3} \right) + 1 = 0.5108256 + 1 = \boxed{1.5108256}.$$

Thus, `free` gets a smaller IDF because it is less rare in the corpus.

Step 4: IDF Values for All Features

Feature	IDF	Feature	IDF
assignment	1.9163	please	1.9163
at	1.9163	prize	1.9163
claim	1.9163	submit	1.9163
click	1.9163	the	1.9163
free	1.5108	tomorrow	1.9163
meeting	1.9163	university	1.9163
money	1.9163	your	1.9163
now	1.9163		

The vocabulary is mostly unique to one document, so most IDF values are identical.

Step 5: Raw TF-IDF for D_1

For

$$D_1 = \text{"free money click now"},$$

each present term occurs once, so the raw term count is 1.

Before normalization:

$$\text{tfidf}(\text{free}, D_1) = 1 \times 1.5108256 = 1.5108256,$$

$$\text{tfidf}(\text{money}, D_1) = 1 \times 1.9162907 = 1.9162907,$$

$$\text{tfidf}(\text{click}, D_1) = 1 \times 1.9162907 = 1.9162907,$$

$$\text{tfidf}(\text{now}, D_1) = 1 \times 1.9162907 = 1.9162907.$$

Raw nonzero vector:

$$[1.9163_{\text{click}}, 1.5108_{\text{free}}, 1.9163_{\text{money}}, 1.9163_{\text{now}}].$$

Step 6: L2 Normalization of D_1

By default, `TfidfVectorizer(norm="l2")` scales each row to unit Euclidean norm.

$$\|D_1\|_2 = \sqrt{1.5108256^2 + 3(1.9162907^2)} \approx 3.6466.$$

Therefore,

$$\text{tfidf}_{\text{norm}}(\text{free}, D_1) = \frac{1.5108256}{3.6466} \approx \boxed{0.414289},$$

$$\text{tfidf}_{\text{norm}}(\text{money}, D_1) = \frac{1.9162907}{3.6466} \approx \boxed{0.525473}.$$

Hence the nonzero values of D_1 are

$$\boxed{\text{click} = 0.525473, \text{free} = 0.414289, \text{money} = 0.525473, \text{now} = 0.525473}.$$

Step 7: TF-IDF for D_2 , D_3 , and D_4

D_2 : meeting tomorrow at university

All four terms have the same IDF, so after L2 normalization:

$$\text{at} = \text{meeting} = \text{tomorrow} = \text{university} = \boxed{0.5}.$$

D_3 : claim your free prize

Same pattern as D_1 :

$$\text{claim} = \text{prize} = \text{your} = 0.525473, \quad \text{free} = 0.414289.$$

D_4 : please submit the assignment

Again all four terms have equal IDF:

$$\text{assignment} = \text{please} = \text{submit} = \text{the} = \boxed{0.5}.$$

Step 8: Final TF-IDF Feature Matrix

Feature order:

[assignment, at, claim, click, free, meeting, money, now, please, prize, submit, the, tomorrow, university, your]

$$X \approx \begin{bmatrix} 0 & 0 & 0 & .5255 & .4143 & 0 & .5255 & .5255 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & .5000 & 0 & 0 & 0 & .5000 & 0 & 0 & 0 & 0 & 0 & 0 & .5000 & .5000 & 0 \\ 0 & 0 & .5255 & 0 & .4143 & 0 & 0 & 0 & 0 & .5255 & 0 & 0 & 0 & 0 & .5255 \\ .5000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & .5000 & 0 & .5000 & .5000 & 0 & 0 & 0 \end{bmatrix}.$$

Thus,

$$X \in \mathbb{R}^{4 \times 15}.$$

Each row is a document; each column is a vocabulary feature; each entry is the normalized TF-IDF weight.